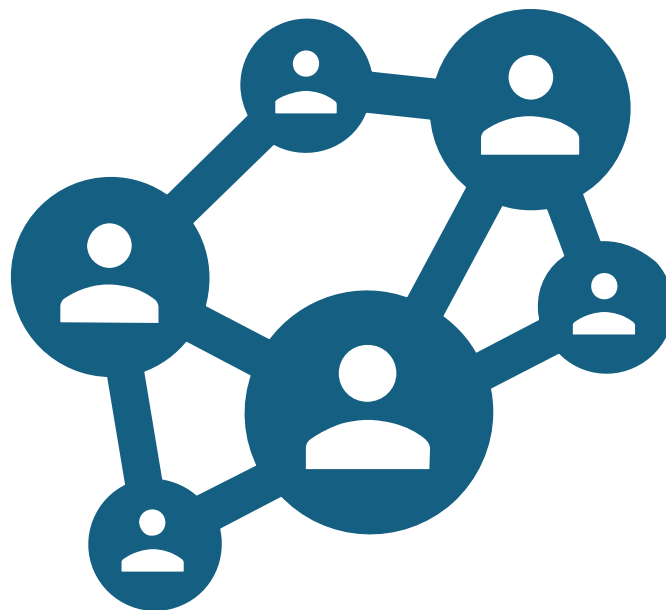


A Network Model of R&D, Competition and Intellectual Property

Bachelor's Degree Final Project

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Description



Firms compete for market share by increasing the quality of their product



Firms can invest in original R&D or reverse-engineer their rival's technology



Companies are embedded in a network, where symmetric links influence both imitation costs and competition levels.



Model simulates a single cycle from an arbitrary starting point until product launch for all firms.

Research Questions

Under what conditions do firms exert effort in original development?

How does patent law infringement affect the quality of products in competitive markets?

Would the elimination of patent protections increase efficiency?

Equations

Product Quality

$$Q_f = Q_0 + x_d + x_e$$

$$Q_0 \geq 0$$

Cost of Development

$$C_d = D \cdot x_d \cdot \left(1 + x_e + \frac{x_d}{2} \right)$$

$$x_d \geq 0$$

Cost of Reverse Engineering (Imitation)

$$C_e = \frac{E}{L_{i,T}} \cdot x_e$$

$$0 < L \leq 1$$

Net Sales

$$S_i = M \cdot \frac{Q_i^2}{Q_i + \sum_{j \neq i}^N L_{ij} \cdot Q_j}$$

Profit

$$P = S - C_d - C_e$$

Parameters

General

Instance

$D=1$

N

$E=1$

Q_0 -Vector

$M=100$

L-Matrix

Scenarios

Single Market	$L=1$
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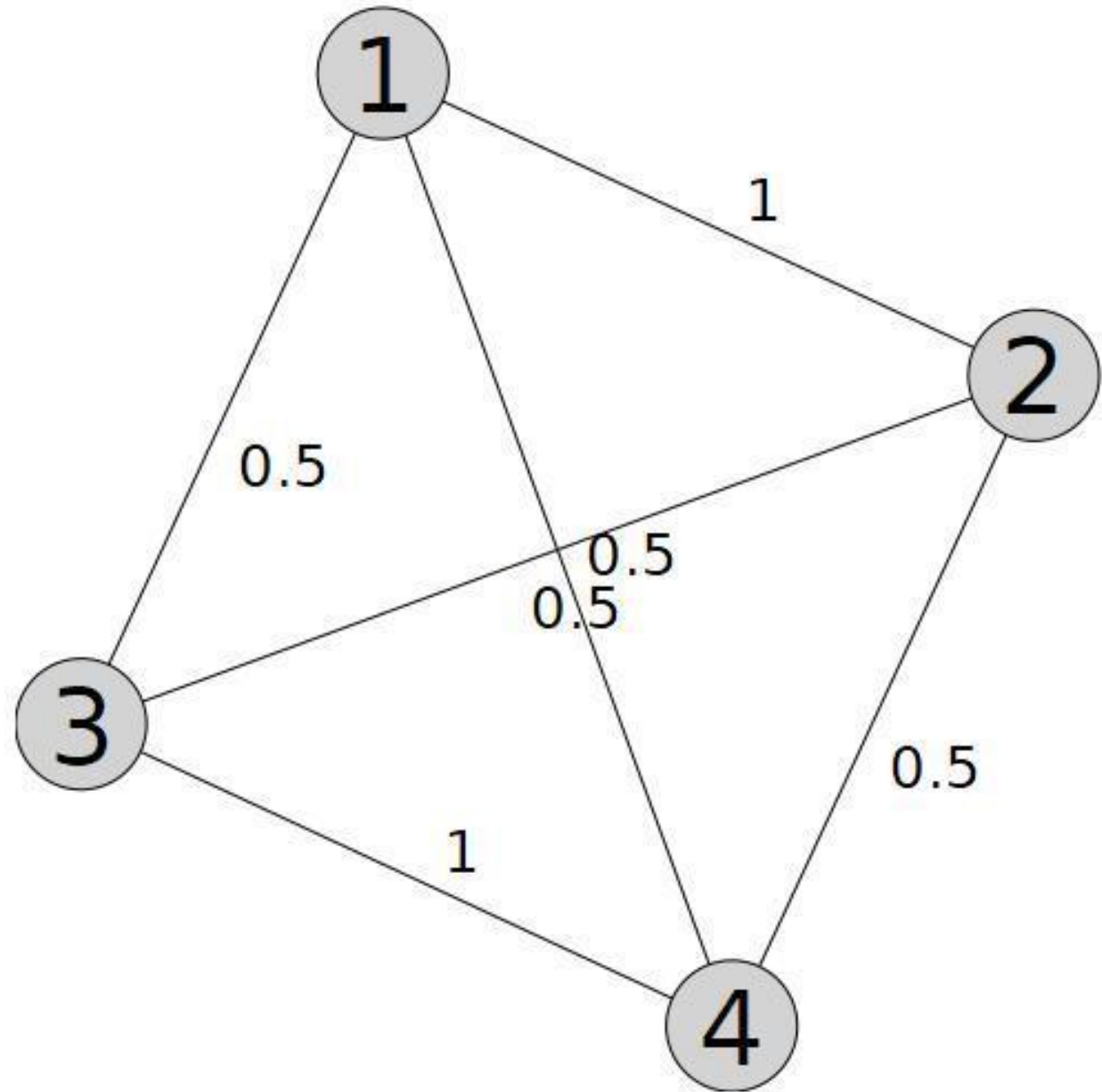
Two Markets	$L=1, L=0.5$
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Closed Silk Road	$L=1, L=0.5, L=0$
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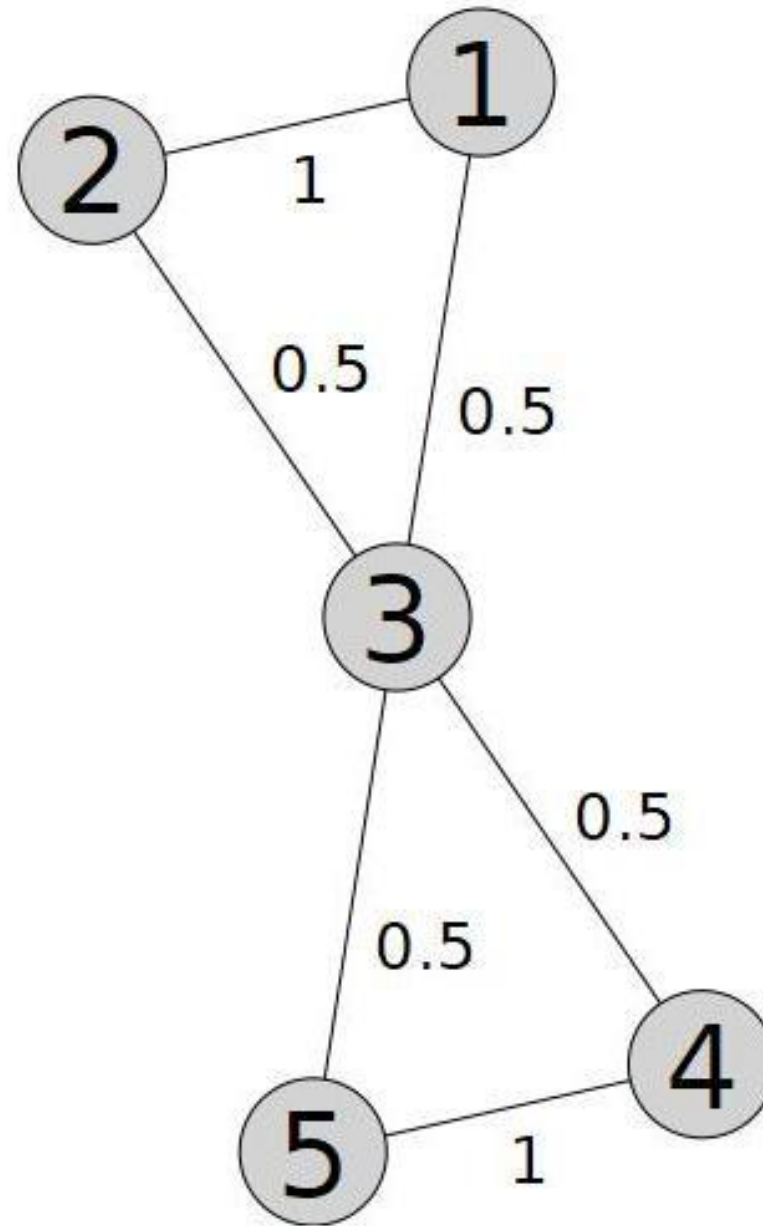
Open Silk Road	$L=1, L=0.5, L=0.2, L=0.05$
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Global Market	$L=0.75$
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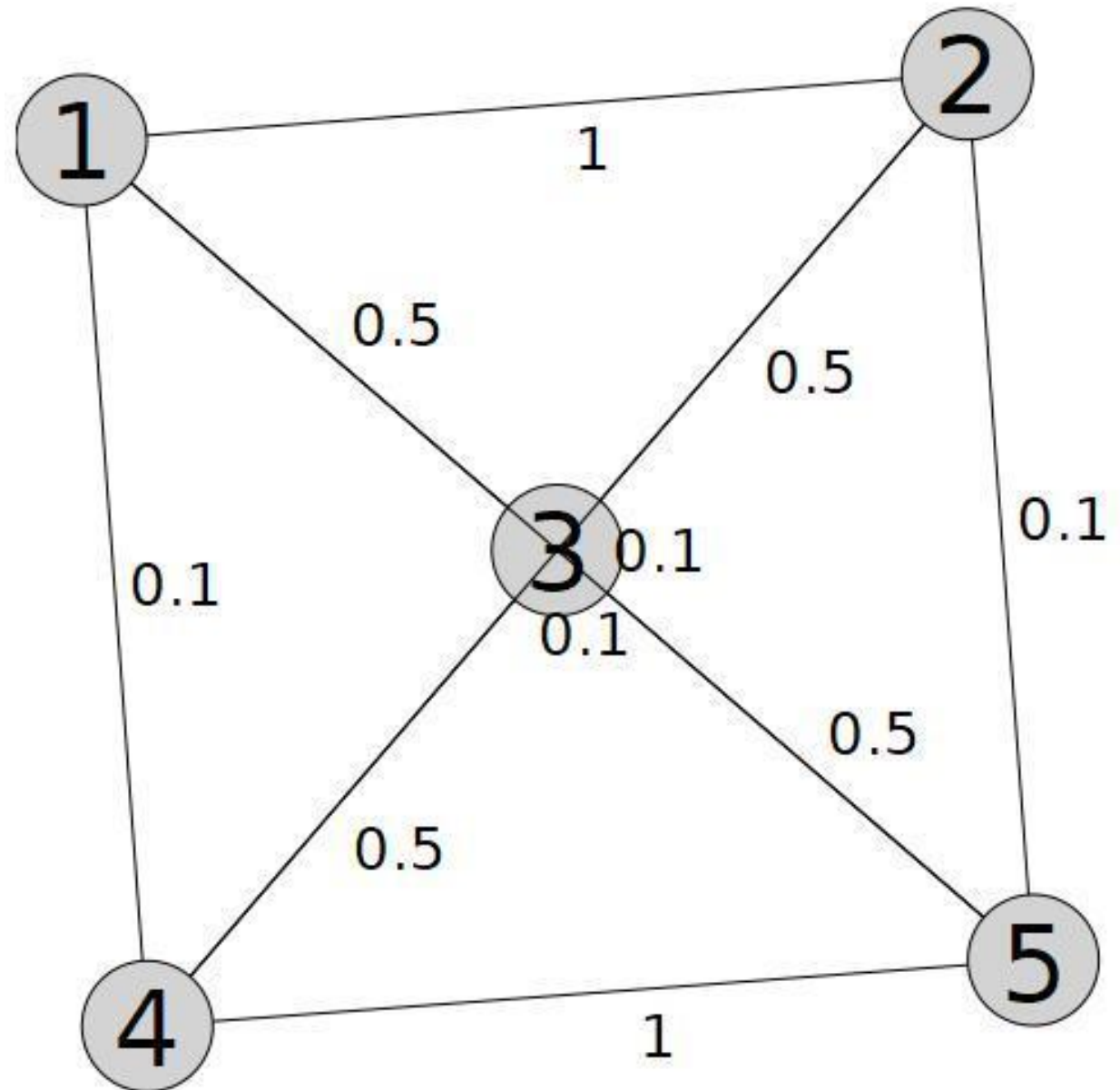
Two Markets



Closed Silk Road



Open Silk Road



Findings: Starting Quality

Model

- Firms starting with an advantage tend to stay ahead
- Firms with low starting qualities struggle to develop in highly competitive markets

Literature



Findings: Competition

Model

- Profits and margins are reduced by increasing competition
- Maximum final qualities are reduced by increasing competition
- Firms can perform well in local markets despite not being the global leader

Literature



Findings: R&D

Model

- Market leaders and Monopolists tend to exert the greatest effort in R&D
- When patent enforcement is lax, non-leading firms tend to exert less R&D effort.

Literature



Findings: Reverse Engineering (Imitation)

Model

- Leaders are unlikely to be harmed by non-competitors copying their technology
- Firms always target the competitors with the highest starting quality
- Firms always reverse-engineer as much as they are allowed

Literature



Findings: Profits

Model

- Reverse Engineering can increase joint profits, especially with market power
- Imitation reduces the distance between leaders and straggling firms

Literature



Findings: Margins

Model

- Firms engaged in original R&D tend to have moderate margins
- Firms relying heavily on imitation tend to achieve wide margins but limited profits

Literature



Answers to Research Questions

Firms exert effort in R&D when they can capture a significant part of the resulting profits.

Market Leaders & Monopolists

Patent law infringement reduces the quality of market leaders by harming their incentives, while it increases the quality of minor firms by allowing them to catch up at little cost.

Quality compression

Reverse engineering can increase efficiency by mitigating cost duplication in R&D. **(See limitations)**

Limitations

R&D cannot be used for strategic differentiation

Simplistic quality demand function

Unrealistic cost curve of reverse engineering

Licensing agreements not modelled

Fusion of the research, patents and goods markets

Deterministic outcomes of research

Exogenous and synchronized product launches across firms

Single launch-cycle simulation